Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/22/2021
PI Name:	Giancardo, Luca Ph.D.		
Project Title:	Actionable Deep Space Stroke Detection with Deep Learning and Retinal Imaging		
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Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
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Space Biology Special Category:	None		
PI Email:	luca.giancardo@uth.tmc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-500-3609
Organization Name:	The University of Texas Health Science Center at Houston		
PI Address 1:	Center for Precision Health, School of Biomedical Informatics (SBMI)		
PI Address 2:	7000 Fannin St, #1200		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2020 TRISH BRASH1901: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Channa, Roomasa M.D. (Baylor College of Medicine) Sheth, Sunil M.D. (The University of Texas Health Science Center at Houston)		
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Task Description:	An untreated stroke event would be destructive for a human deep space exploration mission. Increased cerebrovascular disease risk has been documented after prolonged exposures to ionizing radiations on Earth. Astronauts on deep space exploration missions will be exposed to galactic cosmic rays and solar particles for 30 months, which will lead to accelerated vascular injury likely increasing their risk of stroke, which is exacerbated by the negative effects of microgravity on cerebrovascular autoregulation. On Earth, acute strokes can be successfully treated with anticoagulant or thrombolytic drugs if the event is rapidly diagnosed and the type of stroke (ischemic versus hemorrhagic) is rapidly identified with Computer Tomography (CT) or Magnetic Resonance Imaging (MRI) brain imaging. However, these brain imaging capabilities do not exist in space and alternative robust means of diagnosing and classifying stroke are needed. Due to the homology between retinal and cerebral vessels, and the ease with which retinal images can be acquired non-invasively, retinal images have been studied as a marker for cerebrovascular events. We propose to use a combination of color fundus photos and Optical Coherence Tomography Angiography (OCT-A) images to identify stroke events and stroke type, effectively acting as a proxy for brain imaging. These imaging modalities are non-invasive and deployable in currently existing technologies for deep space missions. We will adapt our automated interpretable image-based deep learning algorithm to identify stroke and stroke type from retinal vascular images, enabling an automated life-saving tool usable on a deep space exploration mission. This approach will leverage the symmetry relationships between the retinal images of each eye in order to identify subtle vasculature changes and at the same time be robust to confounders that affect both eyes at the same time. Significance: If we were able to create a retinal imaging-based quantitative tool to establish the presence of stroke, and stroke type (ischemic versus hemorrhagic), we would be able to indicate the appropriate treatment in a deep space mission stroke emergency, when prompt intervention is of utmost importance in saving astronauts' lives. This is a high-risk high-reward project aiming to create and validate software prototypes towards this goal with a ground-based study which involves a data collection and algorithm development effort. The system proposed has the potential to enable lifesaving treatment in case of a stroke event. Innovation: - We will create an acute stroke database with subjects within a few hours of stroke onset and OCT-A imaging, in addition to neuroimaging, fundus retina images, and clinical assessment. - We will drive innovation by establishing the feasibility of machine learning models to identify acute stroke events and stroke type from retina data, effectively acting as a proxy for brain imaging. - We will be using a first-of-its-kind deep learning model using symmetry-sensitive relationships between the retinal images of each eye. This could enable the algorithm to be robust to space travel induced changes which affect both eyes at the same time, such as Spaceflight Associated Neuro-Ocular Syndrome (SANS). - Our model will be interpretable without having to compromise on specific architecture, as we will be able to study the regions of activation using the epsilon-LRP (Layer-wise Relevance Propagation) algorithm, to understand the image areas responsible for the model decisions. - While some initial work has been done to create machine learning models combining information from fundus images and OCT data, to our knowledge, we will be the first to experiment with a combination of fundus imaging and OCT-A imaging using machine learning approaches. Such combination will capture the optic disc/vasculature with a large field of view (fundus) and the finer information for blow flow (OCT-A).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Long space exploration missions entail the risk of exposing humans to life-threatening medical conditions such as cerebrovascular diseases. Studies on microgravity indicate that cerebrovascular autoregulation is impaired on long duration spaceflight. Other studies have shown increased vascular stiffening, and worsening of atherosclerotic changes due to exposure to ionizing radiation. For example, a study on 86,611 Hiroshima and Nagasaki survivors found a strong association between stroke and radiation exposure, even at very low doses of radiation. The combined continuous low-dose radiations and the microgravity environment expose the astronauts on long spaceflight missions to a significant increase in the risk of a stroke event. Such events can be devastating and we currently do not have a way to prevent them. If we were able to create a retinal imaging-based quantitative tool to establish the presence of stroke and stroke type, we would be able to indicate the appropriate treatment in a deep space mission stroke emergency, when prompt intervention is of utmost importance in saving astronauts' lives. This technology would also have a significant impact on Earth when deployed in mobile stroke units and ambulance, as they will be able to reduce the time of intervention. This is a high-risk high-reward project aiming to create and validate software prototypes towards this goal with a ground-based study which involves a data collection and algorithm development effort. In addition to the significance and impact of the long-term goal described above, the aims of this project will allow to generate new hypotheses and validate existing association about connections between retina microvasculature and acute stroke.
	An untreated stroke event would be destructive for a human deep space exploration mission. Increased cerebrovascular disease risk has been documented after prolonged exposures to ionizing radiations on Earth. Astronauts on deep space exploration missions will be exposed to galactic cosmic rays and solar particles for 30 months, which will lead to accelerated vascular injury likely increasing their risk of stroke, which is exacerbated by the negative effects of microgravity on cerebrovascular autoregulation. On Earth, acute strokes can be successfully treated with anticoagulant or thrombolytic drugs if the event is rapidly diagnosed and the type of stroke (ischemic versus hemorrhagic) is rapidly identified with Computer Tomography (CT) or Magnetic Resonance Imaging (MRI) brain imaging. However, these brain imaging capabilities do not exist in space and alternative robust means of diagnosing and classifying stroke are needed. Due to the homology between retinal and cerebral vessels, and the ease with which retinal images can be acquired non-invasively, retinal images have been studied as a marker for cerebrovascular events. We propose to use a combination of color fundus photos and Optical Coherence Tomography Angiography (OCT-A) images to identify stroke events and stroke type, effectively acting as a

Task Progress:	proxy for brain imaging. These imaging modalities are non-invasive and deployable in currently existing technologies for deep space missions. We will adapt our automated interpretable image-based deep learning algorithm to identify stroke and stroke type from retinal vascular images, enabling an automated life-saving tool usable on a deep space exploration mission. Original project aims/objectives Aim 1: Build a terrestrial acute stroke dataset retinal images and stroke clinical outcomes. Aim 2: Develop and validate an interpretable deep learning model to identify the presence of stroke and stroke type from color fundus photos. Aim 3: Development and validation of interpretable deep learning model to identify presence of stroke and stroke type from OCT-A (and fundus image combination). Project Highlights and Key Findings • Camera acquisition setup completed, data acquisition team trained and 84 subjects recruited. • Preliminary analysis on acute stroke data using OCT-A/OCT with feature engineering indicates that it is possible to detect acute ischemic stroke (AUC 0.89 [CI 0.74-0.99] / AUC 0.86 [CI 0.74-0.96]); however, age is currently a confounder. • Retina vasculature embeddings tested on non-acute stroke data. We found a significant predictive association with our retina vasculature imaging biomarker and stroke subjects (AUC 0.69 [CI 0.65-0.73]). • Identified multiple challenges in the data acquisition that can inform future hardware development. • On Jan 1st, a National Library of Medicine Scholarship was awarded to Ivan Coronado, PhD student working in Dr. Giancardo's Lab, for extending this project by developing algorithms using Adversarial Learning strategies. Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal • Our initial results, while preliminary, support our initial hypothesis that it is possible to detect acute stroke from retina images. • The COVID pandemic did not allow us to start the data acquisition as initially planned. In addition, we found that many acute stroke subjects have issue in focusing onto the camera and/or suffer from droopy eyelids, which does not allow successful imaging. As such our dataset is growing at a slower pace than expected. We will be extending the recruitment period and if this is not sufficient, start acquiring data from non-acute stroke subjects.
Bibliography Type:	Description: (Last Updated: 06/04/2024)
Awards	Coronado I. "National Library of Medicine (NLM) Training Program in Biomedical Informatics and Data Science, to extend the project using algorithms developed, January 2021." Jan-2021
Significant Media Coverage	Giancardo L. "Project featured in Out in Front yearly magazine sent to University of Texas Health donors." Out in Front Magazine, February 2020., Feb-2020